Real-World Applications of Retrieval-Augmented Generation

Healthcare • Finance • Customer Support • Education

Module 2 — Lecture 4

MODULE 2 · LECTURE 4 | DURATION · 60 MIN | SYLLABUS · 2.2 REAL-WORLD APPLICATIONS OF RAG

CENTRAL QUESTION

"The RAG pipeline is the same — but can we design a healthcare RAG system exactly like a customer-support chatbot?"

No. The architecture may be similar, but the domain requirements are very different.



"Technically perhaps. Professionally, absolutely not."

COURSE MAP MODULE 2 · FROM PIPELINE TO PRODUCTION

Where Are We?

LECTURE 1
WHY RAG?

→

LECTURE 2
HOW RAG WORKS

→

LECTURE 3
HOW TO IMPROVE RETRIEVAL

→

LECTURE 4 — TODAY
WHERE DO WE USE RAG?

Healthcare · Finance · Customer Support · Education · Production Design

→

LECTURE 5
BUILD A RAG APPLICATION

Memory transition: Lecture 3 was about retrieval quality. Lecture 4 is about application quality.

BLACKBOARD — KEEP THESE SIX LINES VISIBLE THROUGHOUT THE LECTURE

1 · RAG APPLICATION
= Knowledge + Retrieval + LLM

2 · SAME ARCHITECTURE
≠ Same Risk

3 · SOURCES
Authoritative + Current + Accessible

4 · RESPONSE
Answer + Evidence + Source

5 · BEHAVIOR
Answer OR Abstain OR Escalate

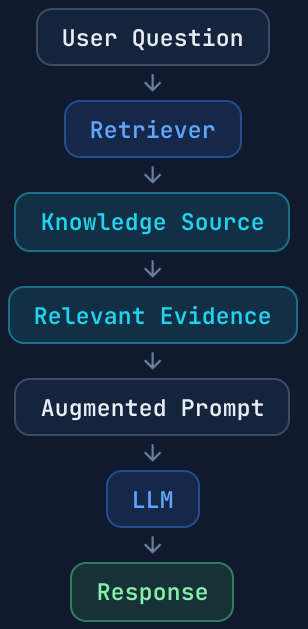
6 · BOUNDARY
RAG retrieves knowledge. It does not automatically execute business processes.

Same engine, different roads — the pattern travels; the obligations are local.

THE REUSABLE PATTERN · ONE PIPELINE · MANY DOMAINS

One Architecture, Many Domains

BASIC RAG REMAINS



BUT THIS CHANGES

- Knowledge Source
- Risk Level
- Access Control
- Freshness Requirement
- Evaluation Criteria
- Human Oversight
- Response Style

Key message: the RAG pattern is reusable; production requirements are domain-specific.

THE ORGANIZATIONAL PROBLEM

Lots of useful knowledge — PDFs · Databases · Policies · Manuals · Circulars · Reports · FAQs · Research — yet employees and customers cannot find answers quickly.

User → Search manually → Open documents → Read pages → Interpret → Answer

THE RAG PROCESS

User Question → Retrieve Relevant Evidence → LLM Synthesizes → Answer + Source

Benefit: RAG converts document search into **evidence-grounded question answering**.

PART A — HEALTHCARE RAG DOMAIN 1 OF 4

Healthcare: Why RAG?

HEALTHCARE INFORMATION CAN BE

- **Large** — years of guidelines and literature
- **Specialized** — deep clinical terminology
- **Frequently updated** — guidance revisions
- **Distributed across documents**
- **Sensitive** — patient and clinical data
- **High consequence** — errors matter

POSSIBLE KNOWLEDGE SOURCES

- Clinical Guidelines
- Hospital Protocols
- Drug Information
- Research Literature
- Patient-Education Material
- Internal SOPs
- Approved FAQs

Important: a healthcare RAG assistant should be designed to support authorized professionals or provide carefully scoped informational assistance — **not blindly replace clinical judgment**.

HEALTHCARE · EXAMPLE 1 AUTHORIZED PROFESSIONAL IN THE LOOP

Clinical Knowledge Assistant

Doctor asks: *"What does our hospital protocol say about managing this procedure?"*

PIPELINE



EXAMPLE RESPONSE STYLE

"According to **Hospital Protocol X, Section 4.2…**"

Key points:

1. …
2. …
3. …

Source: Protocol X, Version 2026.3

RAG advantage: the answer is tied to the **current approved protocol**, rather than only the LLM's general knowledge.

HEALTHCARE · EXAMPLES 2 & 3 RESEARCH ASSISTANT · PATIENT INFORMATION

Research Support & Patient Information

EXAMPLE 2 — RESEARCH ASSISTANT

Researcher asks: *"What recent studies in our approved literature set discuss technique X?"*

Research Papers → Index → Researcher Query → **Retrieve Relevant Papers / Sections** → LLM → **Evidence Summary**

Possible output: major findings · study differences · source papers · publication metadata

Important: the RAG system should preserve the distinction between **what a source says** and **what the model infers**.

EXAMPLE 3 — PATIENT INFORMATION

Patient asks: *"How should I prepare for the procedure scheduled tomorrow?"*

KNOWLEDGE BASE

Approved Patient Preparation Guide · Hospital FAQ · Appointment Instructions

RAG can retrieve the specific approved instructions.

BUT IF THE USER ASKS

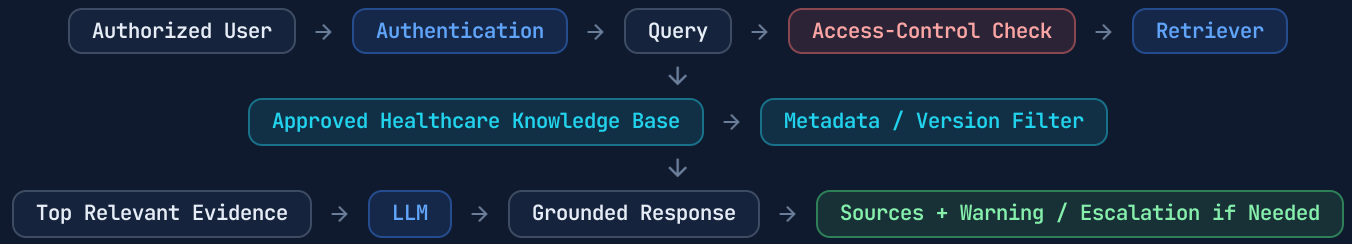
"Should I stop my prescription medication?"

That may require escalation to an appropriate healthcare professional rather than autonomous personalized advice.

📌 **Design principle:** define what the system is allowed to answer and when it must escalate.

HEALTHCARE · ARCHITECTURE IDENTITY BEFORE RETRIEVAL

Healthcare RAG Architecture



WHY ACCESS CONTROL?

Because not every user should see every:

- Patient record
- Internal protocol
- Confidential clinical document

Pattern to remember: identity and permission checks run **before** retrieval, not after it.

HEALTHCARE · DESIGN REQUIREMENTS

SIX OBLIGATIONS BEFORE DEPLOYMENT

What Healthcare RAG Must Emphasize

Source authority

Is the document approved?

Freshness

Is it the latest version?

Privacy

Does retrieval expose sensitive information?

Traceability

Can the answer be verified?

Abstention

Can the system say: "The available evidence does not support an answer"?

Human oversight

When should the user be directed to a qualified professional?

Key statement: in high-stakes RAG, a confident unsupported answer is worse than a careful refusal.

PART B — FINANCE / BANKING RAG DOMAIN 2 OF 4

Why Finance Uses RAG

LARGE KNOWLEDGE COLLECTIONS

- Bank Policies
- Loan Product Documents
- Customer FAQs
- Regulatory Documents
- Internal Procedures
- Annual Reports
- Research Reports
- Compliance Manuals

USERS MAY INCLUDE

- Customers
- Relationship Managers
- Analysts
- Compliance Teams
- Support Agents

One retrieval platform serves very different audiences — **each with its own permissions, vocabulary and risk exposure**. Access control will matter as much as answer quality.

FINANCE · EXAMPLE 1 PRODUCT-SPECIFIC, ALWAYS CURRENT

Loan Product Assistant

Customer asks: *"What documents are required for this home-loan product?"*

✕ WITHOUT RAG

General LLM

↓

May give generic requirements

Trained-in knowledge knows loan documents in general — not **this** product's current policy.

✓ WITH RAG

Question

↓

Retrieve Current Product Policy

↓

Relevant Eligibility / Documentation Section

↓

LLM

↓

Current Product-Specific Answer

POSSIBLE RESPONSE

"According to **Home Loan Product Guide v5.2**, the required documents include…"

Benefit: product changes can be handled by **updating the source documentation** — not retraining anything.

FINANCE · EXAMPLE 2 KNOWLEDGE VS TRANSACTION DATA

Customer-Service Support

Customer: *"My account was debited but the transfer failed."*

RAG RETRIEVES

- Failed Transaction SOP
- Refund Timeline
- Escalation Procedure
- Customer FAQ

THEN THE LLM CAN SYNTHESIZE

- Likely category
- Recommended next step
- Expected timeline
- Escalation channel

IMPORTANT BOUNDARY

RAG can provide **procedure information**, but actual account status should come from **authorized transactional systems / tools** — not be guessed by the language model.

EXCELLENT OPPORTUNITY TO DISTINGUISH

Knowledge retrieval
policies, SOPs, timelines

⇄

Operational transaction lookup
live account state via APIs

Ask: *should RAG guess a customer's current bank balance?* Never — that is a tool call, not a retrieval problem.

FINANCE · EXAMPLES 3 & 4 ANALYST RESEARCH · COMPLIANCE

Research RAG & Compliance RAG

RESEARCH RAG

Analyst asks: *"What did Company X say about revenue growth in the last three annual reports?"*

KNOWLEDGE BASE

2024 Annual Report · 2025 Annual Report · 2026 Annual Report

Query → Retrieve Revenue Sections → Rerank → LLM → Comparative Summary + Sources

An excellent RAG use case because the information exists explicitly in documents.

COMPLIANCE RAG

Employee asks: *"What does our current policy require for transaction X?"*

RAG RETRIEVES

Approved Compliance Manual · Current Regulation · Internal Procedure

Potential response: Requirement · Relevant Clause · Source · Version / Effective Date

CRITICAL REQUIREMENT

Old policy versions must **not silently outrank** the current policy.

Connects to Lecture 3 — metadata: `effective_date · version · status = active`

FINANCE · DESIGN REQUIREMENTS TEN CONSIDERATIONS

What Finance RAG Must Get Right

- Source authority
- Version control
- Effective dates
- Citations
- Audit logs
- User permissions
- Sensitive financial information
- High-quality abstention
- Human escalation
- Separation of factual policy Q&A from personalized financial decisions

Key message: in finance, retrieval correctness and document version are often as important as linguistic quality.

PART C — CUSTOMER SUPPORT RAG

DOMAIN 3 OF 4

Why Support Is a Natural RAG Application

SUPPORT KNOWLEDGE OFTEN ALREADY EXISTS IN

- FAQs
- Product Manuals
- Troubleshooting Guides
- Warranty Policies
- Return Policies
- Known-Issue Databases
- Resolved Support Articles

CUSTOMER QUESTIONS ARE USUALLY

- **Repetitive** — the same questions, every day
- **Document-grounded** — answers live in manuals and policies
- **High volume** — thousands per week

This makes support a **natural RAG use case**.

SUPPORT · EXAMPLE + METADATA PRODUCT IDENTITY MATTERS

Router Troubleshooting & Why Metadata Matters

Customer asks: *"My router's Internet light is red. What should I do?"*

Question → Retrieve Router Model Manual → Retrieve "Red Internet LED" Troubleshooting → LLM → Model-Specific Procedure

Without retrieval: the LLM falls back to generic troubleshooting.

POSSIBLE ANSWER

1. Verify cable connection.
2. Restart the router.
3. Check service-status indicator.
4. If the condition persists, contact support.

Source: Model X Troubleshooting Guide, Section 6

IMAGINE THE COMPANY SELLS

Router X100 | Router X200 | Router Pro

Query: *"How do I reset my router?"* — if the customer owns `model = X200`:

```
product = X200
document_version = latest
language = English
```

Result: the system retrieves instructions for the correct product.

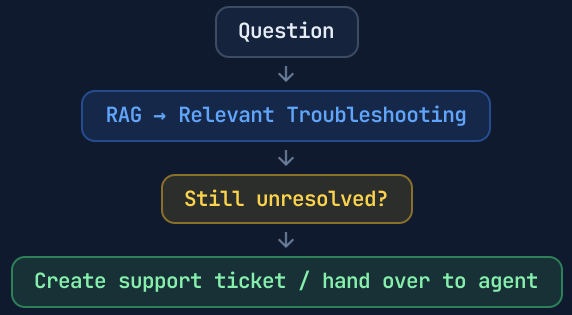
Professional lesson: semantically similar is not enough when the product identity matters.

SUPPORT · ESCALATION & BENEFITS

AUTOMATION WITH A WAY OUT

Escalation Paths & Real Benefits

NOT EVERY PROBLEM SHOULD BE ANSWERED AUTOMATICALLY



A USEFUL ARCHITECTURE — RAG ASSISTANT BRANCHES

- Answer confidently from evidence
- Ask clarifying question
- Escalate to human support

POTENTIAL BENEFITS

- Faster first response
- 24×7 assistance
- Easier product updates
- Reduced repetitive workload for agents
- Consistent answers
- Reduced document-search time
- Source-supported responses

But: poor retrieval can confidently give instructions for the **wrong product**. Therefore retrieval quality remains critical.

Key idea: good automation includes a path out of automation.

PART D — EDUCATION / UNIVERSITY RAG DOMAIN 4 OF 4

Education RAG: Sources & Users

POSSIBLE KNOWLEDGE SOURCES

- Academic Regulations
- Course Syllabus
- Examination Circulars
- Fee Rules
- Admission Policies
- Student Handbook
- LMS Material
- Laboratory Manuals
- Research Repository

USERS

- Students
- Faculty
- Administrators
- Researchers

Ask: **why is a university circular a good RAG source?** It is official, versioned, dated, and answers thousands of repetitive student questions — retrieval keeps every answer attached to the **current** circular.

EDUCATION · THREE ASSISTANTS HELPDESK · COURSE CONTENT · RESEARCH

Three University Assistants

1 · HELPDESK RAG

"Can I submit my examination form after the deadline?"

Question → **Retrieve Examination Rules** → Retrieve Current Circular → LLM → **Answer + Source**

Benefit: changing circulars are updated in the **knowledge source** instead of retraining the model. The same running example students already understand.

2 · COURSE-CONTENT ASSISTANT

"Explain deadlock prevention according to our uploaded Operating Systems notes."

KNOWLEDGE BASE

Course Notes · Textbook Chapters · Faculty PPT · Lab Manual

Student Question → **Retrieve Relevant Teaching Material** → LLM → **Explanation Grounded in Course Material**

Benefit: answers align with **what was actually taught** in the course.

3 · RESEARCH ASSISTANT

"Which papers in our repository use reinforcement learning for cloud anomaly detection?"

Research Repository → **Retrieve Relevant Papers** → Retrieve Relevant Sections → LLM → **Summary + Sources**

Applications: literature navigation · document comparison · paper Q&A · evidence synthesis

Important: the researcher should still **inspect original sources** for important scholarly claims.

EDUCATION · DESIGN CONSIDERATIONS NINE OBLIGATIONS

What a University Assistant Must Get Right

- Correct course / document scope
- Academic-year / version filtering
- Source citations
- Faculty / student access permissions
- No fabrication of academic rules
- Distinguish official policy from teaching material
- Handle conflicting documents
- Update circulars
- Age / access appropriate behavior where applicable

Key statement: a university assistant should know which document is authoritative, not merely which document is semantically similar.

PART E — CROSS-DOMAIN DESIGN

ONE PATTERN · FOUR KNOWLEDGE WORLDS

Same Pipeline, Different Knowledge — Risk Levels Differ

DOMAIN	TYPICAL KNOWLEDGE
Healthcare	guidelines, protocols, approved medical documents
Finance	regulations, policies, reports, product docs
Customer Support	manuals, FAQs, troubleshooting
Education	regulations, syllabus, circulars, course content

BUT ALL FOLLOW

Question → Retrieve Domain Evidence → Augment → Generate

LOWER-RISK

"How do I reset my device?"

Incorrect answer is inconvenient.

MEDIUM-RISK

"What is the university form deadline?"

Incorrect answer may create administrative consequences.

HIGHER-RISK

"What clinical action should be taken?"

Wrong answer can have serious consequences.

THEREFORE

Higher Consequence →
Stronger Source Controls · Stronger Validation · More Human Oversight · More Careful Scope

CROSS-DOMAIN · GOVERNANCE KNOWLEDGE QUALITY BEFORE RETRIEVAL

Authoritative Sources & Version Control

AUTHORITATIVE SOURCES

Ask: *"Should our healthcare RAG ingest every random webpage about medicine?"* **No.**

All Available Information ≠ **Approved Knowledge Base**

KNOWLEDGE GOVERNANCE ASKS

- Who approved this?
- When was it updated?
- Which version is active?
- Who can access it?

Memory line: RAG quality begins before retrieval — with knowledge quality.

FRESHNESS & VERSION CONTROL

Suppose the index contains:

`Policy_v1.pdf` `Policy_v2.pdf` `Policy_v3.pdf`

Which should be retrieved? **Usually: the current active version.**

```
version = 3
effective_date = 2026-10-01
status = active
```

PROBLEM

Without version control, a **semantically perfect but outdated** document may be retrieved.

Connect to Lecture 3: `similarity is not correctness.`

CROSS-DOMAIN · SECURITY

CONSTRAIN RETRIEVAL, NOT JUST DISPLAY

Access Control & Privacy

ACCESS CONTROL

Suppose the knowledge base contains:

- Public FAQ
- Internal Staff Procedure
- Confidential Management Policy
- Private Customer Record

Should every user retrieve everything? **No.**

User → Identity → Permission Check → Allowed Document Set → Retriever

RAG SYSTEMS MAY PROCESS SENSITIVE DATA

personal data · confidential corporate data · financial records · healthcare information · student records

DESIGN QUESTIONS

What data is indexed?
Where is it stored?
Who can retrieve it?
What appears in logs?
How long is it retained?

Key principle: access control should constrain what can be retrieved — not merely what is displayed afterward. A very important production point. (Keep privacy discussion conceptual here.)

CROSS-DOMAIN · TRUST CONTROLS THREE BEHAVIORS EVERY PRODUCTION RAG NEEDS

Cite, Abstain, Escalate

1 · CITATIONS & TRACEABILITY

ANSWER
"The deadline is 18 October."
plus:
SOURCE
Policy X, Section 4.2 · Page 7 · Version 3

WHY

- Verification
- Audit
- Accountability
- User trust
- Debugging

But remind students: a citation is useful only if the cited source actually supports the claim.

2 · ABSTENTION

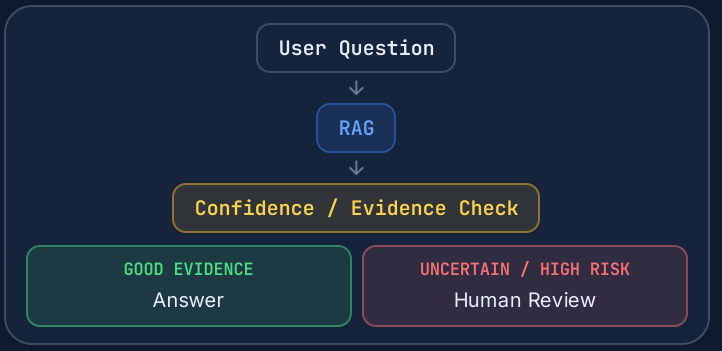
Question: *"What is the penalty for this situation?"* — but the retrieved context does not contain the information.

BAD SYSTEM
invents a penalty.

BETTER SYSTEM
"The available documents do not contain sufficient information to answer this question." Then perhaps: "Please contact the appropriate department."

Principle: the system must be allowed to say, "I don't have sufficient evidence."

3 · HUMAN-IN-THE-LOOP



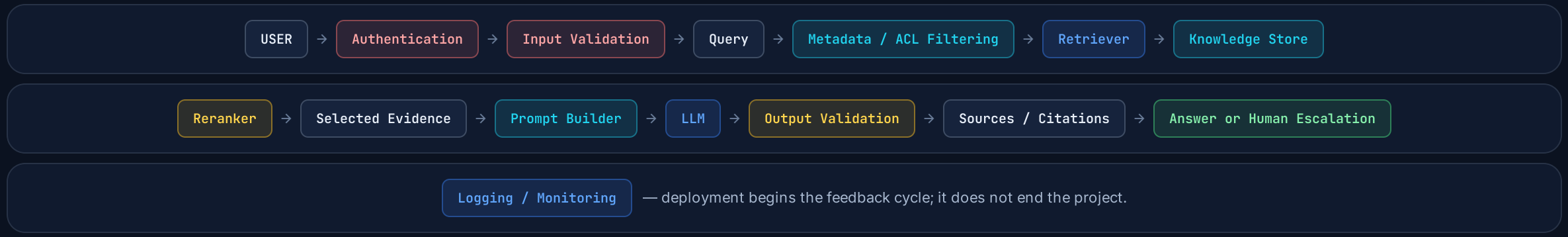
Examples: healthcare → clinician · finance → authorized advisor / compliance team · university → examination office · support → support agent.

Key idea: human escalation is an architectural feature, not evidence that the system failed.

CROSS-DOMAIN · THE FULL SYSTEM

MUCH MORE THAN A VECTOR DATABASE + AN LLM

Production RAG Architecture — Latency & Cost



LATENCY MATTERS TOO — EVERY STAGE ADDS TIME

Query → Embedding → Retrieval → Reranking → Prompt Construction → LLM Inference → Response

A highly accurate RAG system that takes **30 seconds** may still be poor for interactive support. Recall Module 1: `Quality + Latency + Cost + Reliability` all matter.

COST MATTERS — RAG COST DEPENDS ON

embedding generation · vector database · storage · retrieval · reranking · prompt length · LLM tokens · concurrency

More Chunks → Longer Prompt → More Input Tokens → **more cost + latency**. Connect to Lecture 3: `more context is not automatically better context.`

CROSS-DOMAIN · AFTER DEPLOYMENT

MONITOR · THEN COMPARE OBLIGATIONS

What Should We Monitor? Same Tech, Different Obligations

RETRIEVAL

correct sources retrieved? · outdated documents? · empty retrievals?

GENERATION

grounded answers? · unsupported claims? · format compliance?

SYSTEM

latency · error rate · throughput · cost

USERS

unresolved questions · escalation rate · feedback

Key idea: deployment begins the feedback cycle; it does not end the project.

REQUIREMENT	HEALTHCARE	FINANCE	SUPPORT	EDUCATION
Freshness	**Very high**	**Very high**	High	High
Authoritative sources	**Critical**	**Critical**	High	High
Access control	**Critical**	**Critical**	Medium / High	Medium / High
Citation need	High	High	Useful	High
Human escalation	**Critical for high-risk use**	Important	Support agent	Staff / faculty
Wrong-answer impact	**potentially severe**	**potentially severe**	usually operational	academic / admin
Knowledge changes	frequent	frequent	frequent	periodic / frequent

Tell students: "Same RAG technology. Different engineering obligations."

CROSS-DOMAIN · COMBINATIONS

RAG IS ONE COMPONENT OF A SYSTEM

RAG + Systems, RAG + Fine-Tuning

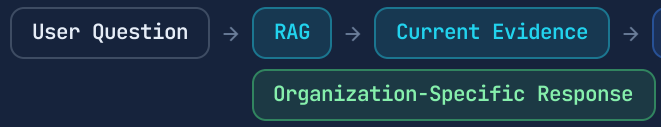
RAG IS NOT ALWAYS THE COMPLETE SOLUTION

- Healthcare → RAG + Clinical System + Human Review
- Banking → RAG + Transactional API + Authentication
- Support → RAG + Ticketing System
- University → RAG + Student Information System

Important distinction: RAG retrieves knowledge. It does not automatically execute business processes. An important production concept — is RAG the same as an operational API? No.

RAG + FINE-TUNING (RECALL MODULE 1)

RAG	Fine-Tuning
changing / external knowledge	learned behavior



Example — university assistant: RAG → current circular, LoRA → fixed Department / Category / Priority schema

Can RAG and fine-tuning be combined? **Yes** — this brings Modules 1 and 2 together.

CLASSROOM ACTIVITY MATCH ARCHITECTURE TO SCENARIO

Application Selection & Five Questions

SCENARIO A

A hospital wants staff to search **approved internal protocols**.

→ Healthcare RAG · source: approved protocols

Critical considerations: version, access, human oversight.

SCENARIO B

A bank wants customers to ask about **current loan-document requirements**.

→ Finance RAG · source: active product policy

SCENARIO C

A router manufacturer wants **automated troubleshooting**.

→ Customer Support RAG · source: model-specific manuals

SCENARIO D

A university wants students to ask about **examination rules**.

→ Education RAG · source: official regulations / circulars

ASK FIVE QUESTIONS BEFORE BUILDING ANY RAG

1. **Who is the user?** Customer? Doctor? Student? Employee?
2. **What knowledge is authoritative?**
3. **How often does it change?**
4. **What happens if the answer is wrong?**
5. **When must the system abstain or escalate?**

These five questions make an **excellent engineering framework** — reuse them in Lecture 5.

RUN AS TRUE / FALSE · DISCUSSION

SEVEN CLAIMS TO TEST INSTINCTS

Common Mistakes

- **FALSE** "Any document can safely be added to the knowledge base." — governance: only approved sources belong in production.
- **FALSE** "The most semantically similar document is automatically authoritative." — similarity is not correctness.
- **FALSE** "RAG eliminates the need for permissions." — access control constrains retrieval itself.
- **TRUE** "If a source changes, we may need to update the index." — freshness is an operational duty, not a one-off.
- **FALSE** "A RAG system should always answer something." — abstention and escalation are valid outputs.
- **FALSE** "Customer-support RAG and clinical RAG can use identical governance." — same architecture ≠ same risk.
- **TRUE** "RAG can be combined with external APIs and fine-tuning." — knowledge + tools + learned behavior.

THE COMPLETE PICTURE MENTAL MODEL · FOUR DOMAINS · ONE PATTERN

Complete Application Mental Model & Final Summary

REAL-WORLD REQUIREMENT → WHO IS ASKING? → WHAT KNOWLEDGE? → IS IT AUTHORITATIVE? → CAN USER ACCESS IT? → RETRIEVE → RERANK / FILTER → GENERATE → VERIFY / CITE → ANSWER OR ESCALATE → MONITOR

Professional takeaway: real-world RAG begins with the application requirement, not with the vector database.

FOUR DOMAINS IN FOUR LINES

Healthcare Ground responses in approved medical / organizational evidence with strong oversight.

Finance Ground responses in current policies, regulations and reports with strict traceability.

Support Retrieve product-specific troubleshooting and policy information quickly.

Education Connect students and staff to current academic regulations, course material and institutional knowledge.

THE ONE COMMON PATTERN

USER NEED → AUTHORITATIVE KNOWLEDGE → RETRIEVE RELEVANT EVIDENCE → LLM SYNTHESIZES → SOURCE-GROUNDED RESPONSE

surrounded by:

Security | Privacy | Freshness | Evaluation | Human Oversight | Monitoring

STUDENTS SHOULD REMEMBER

RAG Application = Good Knowledge + Good Retrieval + Good Generation + Domain-Specific Controls

A good RAG system is not simply technically correct; it must also be appropriate for the risk, users, data and operational requirements of the domain.

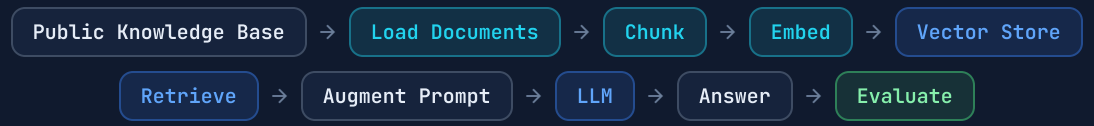
Useful RAG = Right Knowledge + Right Retrieval + Right Controls + Grounded Generation

NEXT LECTURE FROM ARCHITECTURE TO IMPLEMENTATION

Bridge to Lecture 5

"We have now studied why RAG exists, how it works, how retrieval is improved, and where it can be deployed." Now ask: *"Can we actually build one ourselves?"*

LECTURE 5 — PRACTICAL END-TO-END RAG IMPLEMENTATION



RECOMMENDED PRACTICAL SCENARIO — CUSTOMER SUPPORT RAG ASSISTANT

- Easy to understand
- Document-grounded
- Safer for a first implementation
- Clearly demonstrates retrieval
- Students can compare RAG vs non-RAG responses

Closing sentence: "Today we saw where RAG creates value. In Lecture 5, we stop drawing the architecture and build it."